

Ch-4.3

ELEMENTS OF NUMERICAL INTEGRATION

The need often arises for evaluating the definite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain. The basic method involved in approximating $\int_a^b f(x) dx$ is called **numerical quadrature**. It uses a sum $\sum_{i=0}^n a_i f(x_i)$ to approximate $\int_a^b f(x) dx$.

Note that when n is an even integer, the degree of precision is $n + 1$, although the interpolation polynomial is of degree at most n . When n is odd, the degree of precision is only n .

Some of the common **closed Newton-Cotes formulas** with their error terms are listed. Note that in each case the unknown value ξ lies in (a, b) .

$n = 1$: Trapezoidal rule

$$h = \frac{x_1 - x_0}{n} = \frac{1 - 0}{1} = \frac{1}{1} = 1$$

$$x_0 = 0$$

$$\int_{x_0}^{x_1} f(x) dx = \left(\frac{h}{2}\right) [f(x_0) + f(x_1)]$$

$$x_1 = 1$$

$$= \frac{1}{2} [f(0) + f(1)]$$

$n = 2$: Simpson's rule

$$h = \frac{x_2 - x_0}{2}$$

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$= \frac{1 - 0}{2} = \frac{1}{2}$$

$$x_0 = 0, x_1 = \frac{1}{2}, x_2 = 1$$

$n = 3$: Simpson's Three-Eighths rule

$$x_1 = x_0 + h$$

$$= 0 + \frac{1}{3} = \frac{1}{3}$$

$$h = \frac{x_3 - x_0}{3}$$

$$\int_{x_0}^{x_3} f(x) dx = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

$$= \frac{1.5 - 0}{3} = 0.5 = \frac{1}{2}$$

$n = 4$:

$$x_0 = 0$$

$$x_3 = 1.5$$

$$x_1 = x_0 + \frac{1}{2} = 0.5$$

$$x_2 = x_1 + \frac{1}{2} = 0.5 + 0.5 = 1$$

$$\int_{x_0}^{x_4} f(x) dx = \frac{2h}{45} [7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)]$$

Open Newton-Cotes Formulas

The *open Newton-Cotes formulas* do not include the endpoints of $[a, b]$ as nodes. They use the nodes $x_i = x_0 + ih$, for each $i = 0, 1, \dots, n$, where $h = (b - a)/(n + 2)$ and $x_0 = a + h$. This implies that $x_n = b - h$, so we label the endpoints by setting $x_{-1} = a$ and $x_{n+1} = b$, as shown in Figure 4.6. Open formulas contain all the nodes used for the approximation within the open interval (a, b) . The formulas become

$$\int_a^b f(x) dx = \int_{x_{-1}}^{x_{n+1}} f(x) dx \approx \sum_{i=0}^n a_i f(x_i),$$

$$\text{where } a_i = \int_a^b L_i(x) dx.$$

$n = 0$: Midpoint rule

$$h = \frac{x_1 - x_{-1}}{n+2}$$

$$\int_{x_{-1}}^{x_1} f(x) dx = 2hf(x_0)$$

$n = 1$:

$$x_0 = x_{-1} + h$$

$$\int_{x_{-1}}^{x_2} f(x) dx = \frac{3h}{2} [f(x_0) + f(x_1)]$$

if it's a repeating number, write a fraction
or nonterminating number

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$n = 2$:

$$\int_{x_{-1}}^{x_3} f(x) dx = \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)]$$

$n = 3$:

$$\int_{x_{-1}}^{x_4} f(x) dx = \frac{5h}{24} [11f(x_0) + f(x_1) + f(x_2) + 11f(x_3)]$$

Exercise Set 4.3

1. Approximate the following integrals using the Trapezoidal rule. $n=2$

a. $\int_{0.5}^1 x^4 dx$

c. $\int_1^{1.5} x^2 \ln x dx$ $h = \frac{1.5 - 1}{1} = \frac{0.5}{1} = 0.5$

e. $\int_1^{1.6} \frac{2x}{x^2 - 4} dx$

g. $\int_0^{\pi/4} x \sin x dx$

b. $\int_0^{0.5} \frac{2}{x - 4} dx$

d. $\int_0^1 x^2 e^{-x} dx$

f. $\int_0^{0.35} \frac{2}{x^2 - 4} dx$

h. $\int_0^{\pi/4} e^{3x} \sin 2x dx$



c. $\int_1^{1.5} x^2 \ln x \, dx$ $n=1$

$$h = \frac{1.5 - 1}{1} = 0.5$$

$$\frac{0.5}{2} [1.5^2 \ln 1.5 + 1^2 \ln 1] = 0.2281$$

$$\int_1^{1.5} x^2 \ln x \, dx = 0.1923$$

↙ subtract to find error bound

5.

Repeat Exercise 1 using Simpson's rule. $n=2$

$$\text{c. } \int_1^{1.5} x^2 \ln x \, dx = \frac{0.25}{3} \left[1^2 \ln 1 + 4(1.25^2 \ln 1.25) + (1.5^2 \ln 1.5) \right]$$
$$= 0.19224$$

$$h = \frac{1.5-1}{2} = 0.25$$

$$x_0 = 1$$

$$x_1 = 1 + 0.25 = 1.25$$

$$x_2 = 1.5$$

open
Cotes

$n=0$ midpoint

$n=1$

$n=2$

$n=3$

closed
Cotes

$n=1$ trapezoidal

$n=2$ Simpson

$n=3$ Simpson $\frac{1}{3}$

$n=4$

open Cotes

$$h = \frac{1.5-1}{0+2}$$

9.

Repeat Exercise 1 using the Midpoint rule. $n=0$

c. $\int_{1=x_{-1}}^{1.5=x_1} x^2 \ln x \, dx$

$x_0 = 1 + 0.25 = 1.25$

$$h = \frac{1.5-1}{0+2} = \underline{0.25}$$

$$= 2(0.25)(1.25^2 \ln 1.25) = \underline{0.1743}$$

$$2h \times f(x_0)$$

$$\underline{0.1923}$$

$$x_0 = x_{-1} + h$$

The Trapezoidal Rule

To derive the Trapezoidal rule for approximating $\int_a^b f(x) dx$, let $x_0 = a$, $x_1 = b$, $h = b - a$ and use the linear Lagrange polynomial:

$$\int_a^b f(x) dx = \frac{h}{2}[f(x_0) + f(x_1)]$$

Simpson's Rule

Simpson's rule results from integrating over $[a, b]$ the second Lagrange polynomial equally-spaced nodes $x_0 = a$, $x_2 = b$, and $x_1 = a + h$, where $h = (b - a)/2$.

$$\int_{x_0}^{x_2} f(x) \, dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$n = 0$: Midpoint rule

$$\int_{x_{-1}}^{x_1} f(x) \, dx = 2h f(x_0)$$

h. $\int_0^{\pi/4} e^{3x} \sin 2x \, dx$

trapezoidal method

$$n=1$$

$$h = \frac{\pi/4 - 0}{1} = \frac{\pi}{4}$$

exact 81.359

$$= \frac{h}{2} [f(x_0) + f(x_1)]$$

$$= \frac{\pi/4}{2} \left[\cancel{e^{3(0)} \sin(2(0))} + e^{3(\pi/4)} \sin 2(\pi/4) \right]$$

$$= \frac{\pi}{8} \left[e^{3(\pi/4)} \sin(2 \times \pi/4) \right]$$

3.

Find a bound for the error in Exercise 1 using the error formula, and compare this to the actual error.

$$\int_{x_0=0.5}^{x_1=1} x^4 dx = \frac{h}{2} [f(x_0) + f(x_1)] = \frac{0.5}{2} [0.5^4 + 1^4] = 0.265625$$

$$h = \frac{1 - 0.5}{1} = 0.5$$

$$\int_{0.5}^1 x^4 dx = \left[\frac{x^5}{5} \right]_{0.5}^1 = \left[\frac{1^5}{5} - \frac{0.5^5}{5} \right] = 0.19375$$